

Company Name: D-Wave Quantum Inc

Company tagline: Quantum Realized

Company Overview: D-Wave is a leader in the development and delivery of quantum computing systems, software, and services. It is the world's first commercial supplier of quantum computers, and the first and only to offer dual-platform quantum computing products and services, spanning both annealing and gate-model quantum computing technologies. D-Wave's mission is to help customers realize the value of quantum today through enterprise-grade systems available on-premises and via its Leap™ quantum cloud service, which offers 99.9% availability and uptime. More than 100 organizations across commercial, government and research sectors trust D-Wave to address complex computational challenges using quantum computing.

D-Wave is a full-stack quantum computing company, building and delivering annealing and gate-model quantum computers that will address the full range of complex computational problems for customers in business, research, and the public sector. Through cloud services, application development tools, and professional services, D-Wave solutions support the end-to-end quantum computing journey.

Founded in 1999, D-Wave is the leader in the development and delivery of quantum computing systems, software, and services and is the world's first commercial supplier of quantum computers. Our employees are experts in their fields across processor development, chip fabrication, physics, cloud infrastructure, and more. Our company has a shared passion and a simple mission: to unlock the power of quantum computing to help solve the world's most challenging problems. To do so, we've brought together a leadership team with unparalleled experience in business, science, and technology. The team comprises seasoned technology leaders and distinguished domain experts with decades of experience in their respective fields.

Core Focus Area: Quantum Computing

Headquarters: Palo Alto, California, USA

Founded: 1999

Website: www.dwavequantum.com

Problem: Real World Knapsack Problem

Motivation: The knapsack problem is a well-known academic problem that has many important real-world applications. For example, perhaps a retail company needs to load containers with items so that it has the highest total value (in order to maximize profits) while respecting the containers size and weight requirements. This important optimization problem can find its roots in the knapsack problem.

Impact Statement: Even small improvements to this hard optimization problem can have large impacts on a company's performance metrics like profits and losses.

Detailed Problem Description: Build a real-world application demo of the knapsack problem targeted towards a specific problem in a specific industry. For example, the container loading problem in the retail industry referenced in the "Motivation" section. You may choose any application in any industry, so long as the foundational, underlying problem is the knapsack problem (or a variation of it).

The problem must include, at a minimum:

- Inputs: list of items with value and weight, weight restriction
- Objective: maximize value of items selected
- Constraint: total weight of items selected less than or equal to weight restriction

Additional inputs and constraints can be added as desired for a more realistic scenario.

To get everyone started, a small basic example can be found at <https://github.com/dwave-examples/knapsack>.

Requirements:

- Should run using D-Wave's Ocean SDK
- Include user-friendly interface
- Must include a sample dataset for testing

Prerequisites:

- Python programming

Key Challenges:

1. *Research real-world requirements:* Students will be required to research real-world requirements of their selected knapsack application problem to make a realistic demo.

2. *Map real-world requirements to model:* Students will be required to map real-world constraints and requirements to a mathematical model that can be incorporated into a Python demo.
3. *Build a user-friendly interface:* Students must determine how to build a demo that non-technical users from industry can easily interact with and understand.

Resources and References:

- Documentation: <https://docs.dwavequantum.com>
- GitHub Example Repository: <https://github.com/dwave-examples/knapsack>
- Ocean SDK Repository: <https://github.com/dwavesystems>
- Cloud Platform: <https://cloud.dwavesys.com/leap/>